COMP3018 Probabilistic Machine Learning

Artefact 1 Assessment Rubric

	High Distinction	Distinction	Credit	Pass	Fail
Technical Report (27%)					
Individual Component	Outstanding and comprehensive report. Excellent analysis of dataset uncertainty. Deep, critical evaluation and comparison of Frequentist and Bayesian approaches. Stakeholder conflicting requirements are carefully analysed and resolved.	Strong, well-structured report. Clear, accurate comparison of Frequentist and Bayesian methods. Strong dataset exploratory analysis and logical justification of the selected models and technical trade-offs.	Well-written report covering all required sections. Adequate explanation of Frequentist and Bayesian methods. Identifies stakeholder conflicts and dataset features, but lacks deep critical insight.	Satisfactory report. Basic concepts of probability are present. Minimal or surface-level comparison of Frequentist/Bayesian approaches. Stakeholder analysis or dataset exploration is brief or incomplete.	Poorly written or incomplete report. Fails to understand probabilistic foundations. Missing substantive comparison of models or ignores human factors/stakeholder requirements.
Technical Implementation (30%)					
Individual Component	Sophisticated, robust probabilistic architecture. Code is of an excellent standard, highly automated, and fully implements both Frequentist and Bayesian estimation pipelines to handle dataset uncertainty.	Well-designed implementation. Strong use of statistical libraries. Models accurately ingest data and estimate parameters using both approaches. Code is functional and follows best practices.	Logical implementation meeting core requirements. Models function and generate basic point/distribution estimates, but may lack computational efficiency or elegant structural design.	Meets minimum requirements. Generates basic outputs but struggles with Bayesian integration, exhibits poor code structure, or relies on manual data processing steps.	Architecture is insufficient or incomplete. Code is nonfunctional or fails to implement the required probabilistic estimation models.

Student Name:	Overall Mark:	Grade:

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Video Demonstration (5%)					
Individual Component	Highly professional, clear demonstration within the 3-5 minute limit. Excellent technical quality, providing detailed insights into pipeline execution and a sharp comparison of Frequentist point estimates vs. Bayesian intervals.	Clear, well-presented demonstration. Good structure and technical quality. Adheres to the time limit and effectively compares the probabilistic outputs of both models.	Demonstrates core functionality but presentation might lack clarity in explaining the probabilistic logic or output differences. Acceptable technical quality.	Demonstration is difficult to follow, exceeds time limits, or doesn't clearly show the probabilistic models handling uncertainty. Poor technical quality.	Missing, incomplete, or unclear. Does not show a working probabilistic estimation system
Professional Standards (5%)					
Individual Component	Exemplary standard of academic writing. Adherence to the 1,500-word limit, formatting, IEEE citation, and AI policy guidelines (Full AI declaration and appendix included). Self-Assessment included.	Professional writing. Minor errors in formatting, word count, submission, selfassessment, or citation. AI usage is correctly declared.	Satisfactory standard of writing. Understandable but with several errors in formatting, submission guidelines, or incomplete AI logs.	The report is difficult to read or understand. Significant deviations from submission requirements, word counts, or minimal AI policy adherence.	Does not meet professional standards. Ignores submission instructions, academic standards, or is missing required AI declarations.

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Project Management and Workflow (13%)			.		.
Team Component	Excellent project management. Extremely wellstructured A1_YourFAN directory, complete README and requirements files, and clear evidence of proactive task planning for the data ingestion pipeline.	Strong project management. Well-organised submission structure, logical technical decisions, and clear evidence of good planning. Includes required system files.	Satisfactory project management. Meets basic structural requirements with adequate planning for the overarching data pipeline.	Minimal project management. Disorganised file structure, missing requirements.txt, or lack of logical workflow	No evidence of project management. Fails to follow directory requirements and lacks structural organisation..
Team Collaboration and Integration (20%)					
Team Component	Excellent Integration Statement (under 200 words). Deep, insightful analysis of how the individual's data subset and estimation module seamlessly aligns with the overarching	Clear and comprehensive Integration Statement. Strong alignment between the individual module's probabilistic outputs, team objectives, and system governance.	Adequate Integration Statement. Outlines the individual role but lacks deep analysis of data contracts and how the module integrates with the broader team framework.	Basic integration statement. Mentions the team context but provides vague or insufficient explanation of module integration and the overarching system.	Missing or inadequate integration statement. No evidence of team alignment or understanding of the broader probabilistic system.

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	probabilistic network architecture.				

Self-Assessment

- Individual Component (Technical Report): I think I hit the HD mark well except for the Bayesian vs Frequentist comparison, I could have gone more in-depth (which is done in the notebook), but I was struggling to adhere to the word limit. Stakeholder conflicting requirements are addresses well.
- Individual Component (Technical Implementation): Code is consistent good quality with comments and is modular. Automates the entire process from end to end.
- Individual Component (Video): Video covers everything and is within time limit (just) but is not of high production standard.
- Individual Component (Professional Standards): Word limit, reference format, general formatting, AI usage, and self-assessment are all completed and done to standard.
- Team Component (Project Management and Workflow): All team/project standards are followed exactly as detailed in the task sheet. README.md file gives exact instructions and requirements.txt includes all dependencies.
- Team Component (Team Collaboration and Integration): Covers the overall project and goes into detail how my component fits in.